

Metal 3D Printing in Aerospace: How Additive Manufacturing Is Reinventing Aircraft Components

From jet engine brackets to satellite structures, metal additive manufacturing is reshaping what is possible in aerospace design and production.

By Joshua U. Otaigbe, PhD | April 28, 2026 | 7 min read

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Introduction: Manufacturing's Quiet Revolution

For most of aviation's history, if you wanted to make a metal aircraft part, you started with a large block of metal and machined away everything you did not need. For a complex aerospace component, this could mean removing 80 to 90 percent of the original material as chips and shavings. It is an expensive, time-consuming process that places hard limits on what shapes are possible to produce.

Metal additive manufacturing, commonly known as metal 3D printing, turns this approach upside down. Instead of removing material, it builds parts up layer by layer from metal powder or wire, adding only what is needed. This fundamental shift in how parts are made is enabling designs that were previously impossible to manufacture, dramatically reducing material waste, and opening up new possibilities for rapid iteration and customization in aerospace.

How Metal 3D Printing Works

Several different additive manufacturing processes are being used in aerospace, each with different capabilities and best applications. The two most widely adopted for structural metal parts are powder bed fusion and directed energy deposition.

Powder Bed Fusion

In powder bed fusion, a thin layer of metal powder is spread across a build platform. A high-powered laser or electron beam then selectively melts the powder according to the part's digital design file. The platform drops slightly, another layer of powder is spread, and the process repeats, building the part up layer by layer. The result is an extremely precise, fully dense metal

part that can have internal features, channels, and lattice structures that no traditional machining process could produce.

Directed Energy Deposition

This process uses a focused energy source, typically a laser, to melt metal powder or wire as it is deposited onto a substrate. It is particularly useful for building up or repairing existing parts, and for producing large structural components that would be too big for a powder bed system. Some aerospace companies are using directed energy deposition to restore worn turbine blades, extending their service life rather than replacing them.

The Design Freedom Advantage

The most transformative aspect of metal additive manufacturing is not the process itself but the design freedom it enables. When engineers know that any shape can be manufactured, they stop designing for manufacturability and start designing purely for performance. This unleashes a set of design strategies that are simply not possible with conventional manufacturing.

"Additive manufacturing does not just change how we make parts. It changes what parts we can imagine making."

Topology Optimization

Using sophisticated computer algorithms, engineers can specify the loads a part must carry and let the software determine the optimal material distribution to carry those loads. The resulting designs often look organic, resembling bones or coral rather than conventional machined parts. They use material only where it is structurally needed, achieving maximum strength with minimum weight. Topology-optimized brackets and structural fittings printed in titanium have achieved weight reductions of 40 to 60 percent compared to their conventionally manufactured predecessors.

Internal Cooling Channels

Jet engine components operate at temperatures that would melt many materials. Managing this heat requires sophisticated internal cooling systems that circulate cool air through channels inside the hot metal parts. Additive manufacturing allows engineers to create cooling channels with complex, optimized geometries that were previously impossible to machine. The result is better cooling with less weight and fewer parts.

Part Consolidation

Complex assemblies that previously required dozens of individual parts fastened together can often be redesigned as a single printed component. This eliminates the weight of fasteners, reduces the risk of joint failures, and simplifies assembly. GE Aviation famously consolidated a fuel nozzle for the LEAP jet engine from 20 separate parts into a single 3D-printed component that is 25 percent lighter and five times more durable than its predecessor.

Materials at the Core

The aerospace industry's demanding performance requirements have driven the development of an impressive portfolio of printable metal alloys. Each material family brings distinct advantages to specific applications.

Material	Key Properties	Typical Applications
Titanium (Ti-6Al-4V)	High strength-to-weight ratio, corrosion resistance	Structural brackets, airframe fittings, medical implants
Inconel 718	Excellent high-temperature strength	Turbine blades, combustor components, exhaust systems
Aluminum (AlSi10Mg)	Lightweight, good thermal properties	Housings, brackets, heat exchangers
17-4 PH Stainless	High strength, good corrosion resistance	Fittings, fasteners, tooling components
Cobalt-Chrome	Extreme wear and heat resistance	Turbine components, bearings, cutting tools

Certification: The Biggest Hurdle

Despite its remarkable capabilities, metal additive manufacturing faces a significant challenge in aerospace: certification. Aviation regulators require extensive documentation and testing before any new manufacturing process can be used for flight-critical parts. This is not bureaucratic obstruction; it reflects the industry's commitment to safety. But it does mean that the path from a promising new additive technology to a certified, production-ready process can take years.

The core challenge is process variability. Unlike machining, where the properties of the finished part are determined primarily by the starting material, additive manufacturing properties depend on a complex interplay of laser power, scan speed, powder characteristics, build orientation, and post-processing treatments. Demonstrating that these variables are sufficiently controlled to produce consistent, reliable parts is the central challenge in additive manufacturing certification.

Key certification requirements include:

- Comprehensive characterization of the powder feedstock and its lot-to-lot variability

- Qualification of specific machine and parameter combinations for each alloy and part geometry
- Non-destructive inspection methods validated for additive-manufactured microstructures
- Fatigue and fracture testing that accounts for surface roughness and internal porosity
- In-process monitoring systems that detect and flag anomalies during build

The Road Ahead

The aerospace additive manufacturing market is growing rapidly, driven by the clear performance and efficiency advantages of the technology. Major engine manufacturers, airframers, and defense contractors are all investing heavily in additive capabilities. As certification frameworks mature and the technology matures, expect to see additive-manufactured components move from non-critical brackets and housings toward primary structural applications.

At Flaney Associates, we help aerospace companies develop and qualify additive manufacturing processes, from alloy selection and process parameter development to microstructural characterization and certification testing support. Our team understands both the technical and regulatory dimensions of bringing additive parts to flight.

For more information or if you have any questions, please contact the author:

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